New Wheels Project

Problem Statement

Business Context

A lot of people in the world share a common desire: to own a vehicle. A car or an automobile is seen as an object that gives the freedom of mobility. Many now prefer pre-owned vehicles because they come at an affordable cost, but at the same time, they are also concerned about whether the after-sales service provided by the resale vendors is as good as the care you may get from the actual manufacturers.

New-Wheels, a vehicle resale company, has launched an app with an end-to-end service from listing the vehicle on the platform to shipping it to the customer's location. This app also captures the overall after-sales feedback given by the customer.

Objective

New-Wheels sales have been dipping steadily in the past year, and due to the critical customer feedback and ratings online, there has been a drop in new customers every quarter, which is concerning to the business. The CEO of the company now wants a quarterly report with all the key metrics sent to him so he can assess the health of the business and make the necessary decisions.

As a data analyst, you see that there is an array of questions that are being asked at the leadership level that need to be answered using data. Import the dump file that contains various tables that are present in the database. Use the data to answer the questions posed and create a quarterly business report for the CEO.

Business Questions

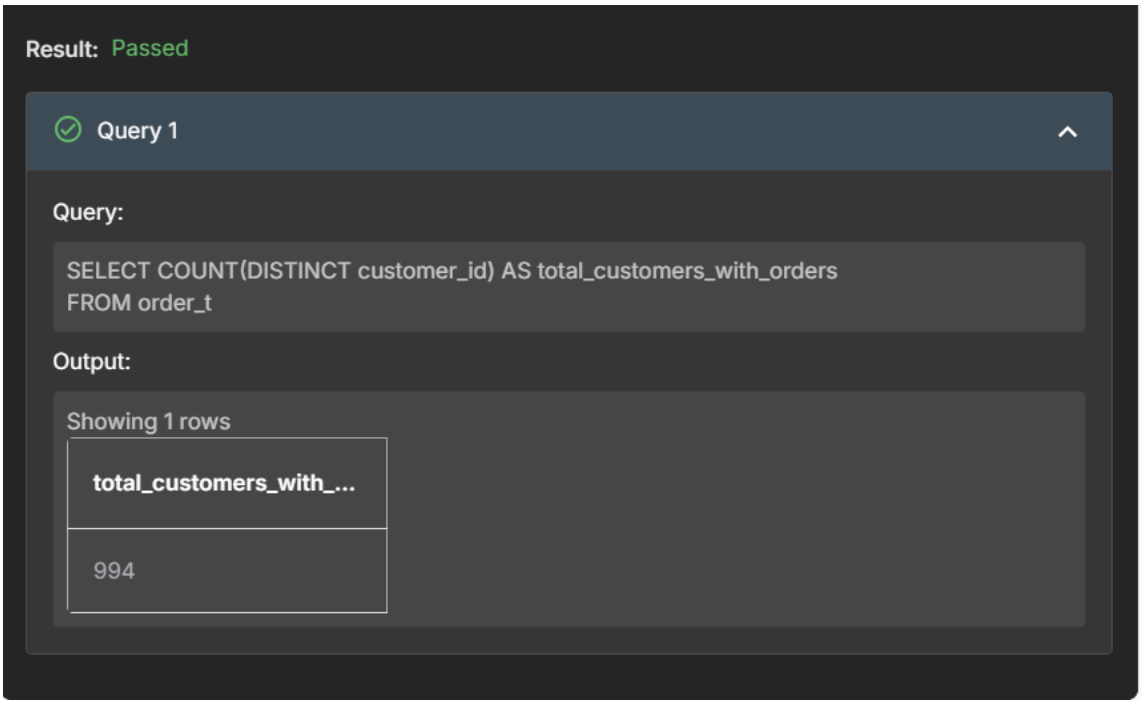
Question 1: Find the total number of customers who have placed orders. What is the distribution of the customers across states?

1.1 Find the total number of customers who have placed orders.

Solution Query:

```
SELECT COUNT(DISTINCT customer_id) AS total_customers_with_orders
FROM order_t;
```

Output:



Observations and Insights:

- The result of the query is **994**, meaning there are **994 distinct customers** with orders.
- A total of **994 unique customers** has interacted with the system by placing orders, which provides a healthy level of customer engagement.

1.2 What is the distribution of the customers across states?

Solution Query:

```
SELECT c.state, COUNT(DISTINCT o.customer_id) AS customer_count
FROM order_t o
JOIN customer_t c USING (customer_id)
GROUP BY c.state
ORDER BY customer_count DESC;
```

Output:

Result: Passed

Query 1

Query:

```
SELECT c.state, COUNT(DISTINCT o.customer_id) AS customer_count
FROM order_t o
JOIN customer_t c USING (customer_id)
GROUP BY c.state
ORDER BY customer_count DESC
```

Output:

Showing first 10 rows out of 49 rows

state	customer_count
Texas	97
California	97
Florida	86
New York	69
District of Columbia	35
Ohio	33
Colorado	33

Observations and Insights:

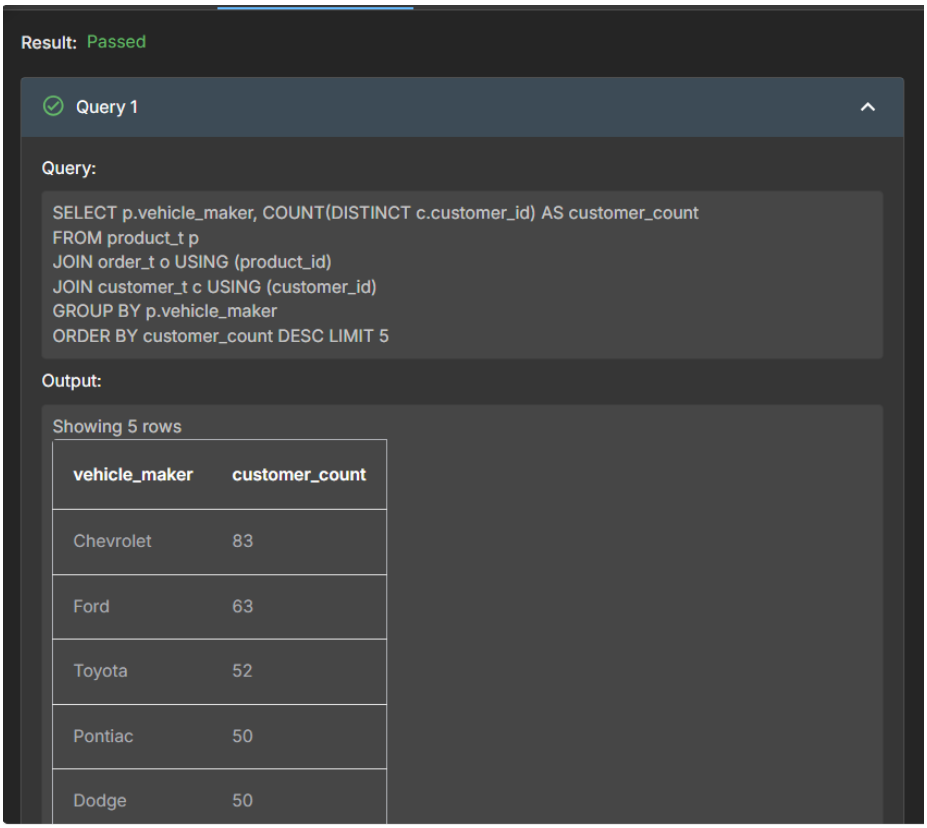
- ***Texas*** and ***California*** are tied as the states with the highest number of distinct customers,
- **Florida** and **New York** Also show significant customer bases, with **86** and **69** customers respectively.

Question 2: Which are the top 5 vehicle makers preferred by the customers?

Solution Query:

```
SELECT p.vehicle_maker, COUNT(DISTINCT c.customer_id) AS customer_count
FROM product_t p
JOIN order_t o USING (product_id)
JOIN customer_t c USING (customer_id)
GROUP BY p.vehicle_maker
ORDER BY customer_count DESC LIMIT 5;
```

Output:



vehicle_maker	customer_count
Chevrolet	83
Ford	63
Toyota	52
Pontiac	50
Dodge	50

Observations and Insights:

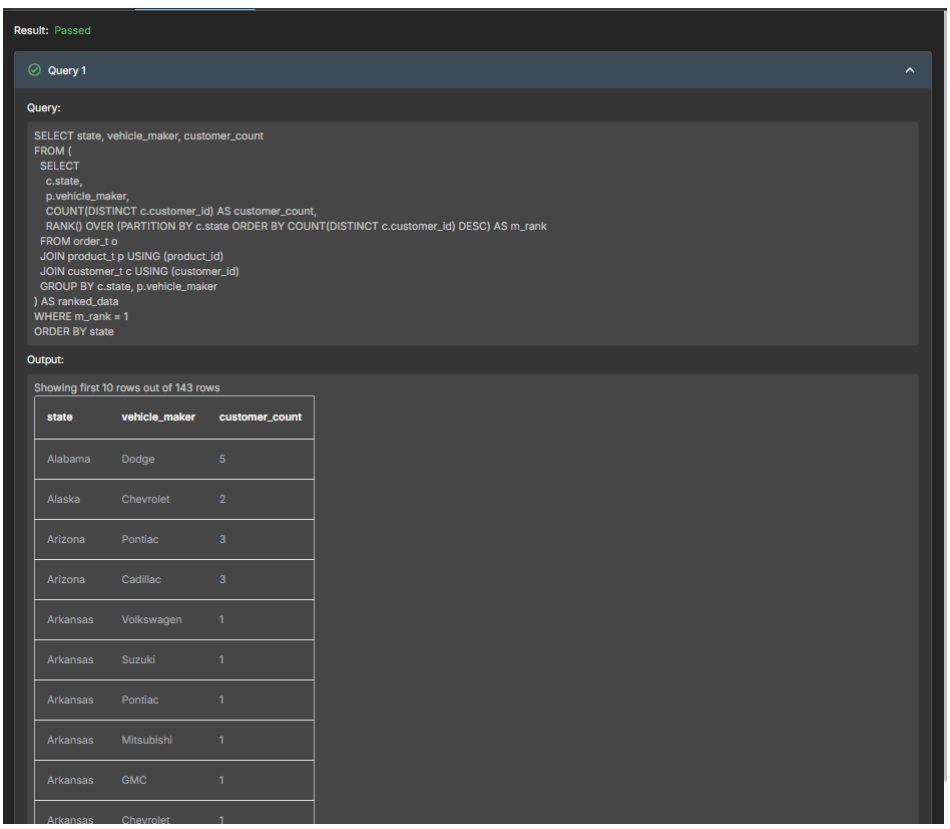
- **Chevrolet** Leads the Market with **83 customers**, Chevrolet is the most preferred brand.
- **Ford and Toyota** maintain a strong mid-tier presence, reflecting possibly good pricing, reliability, or regional availability.
- **Pontiac** and **Dodge,** even though they have the same number of customers, might be popular with a specific group of people or have loyal buyers.

Question 3: Which is the most preferred vehicle maker in each state?

Solution Query:

```
SELECT state, vehicle_maker, customer_count
FROM (
  SELECT
    c.state,
    p.vehicle_maker,
    COUNT(DISTINCT c.customer_id) AS customer_count,
    RANK() OVER (PARTITION BY c.state ORDER BY COUNT(DISTINCT c.customer_id)
DESC) AS m_rank
  FROM order_t o
  JOIN product_t p USING (product_id)
  JOIN customer_t c USING (customer_id)
  GROUP BY c.state, p.vehicle_maker
) AS ranked_data
WHERE m_rank = 1
ORDER BY state;
```

Output:



state	vehicle_maker	customer_count
Alabama	Dodge	5
Alaska	Chevrolet	2
Arizona	Pontiac	3
Arizona	Cadillac	3
Arkansas	Volkswagen	1
Arkansas	Suzuki	1
Arkansas	Pontiac	1
Arkansas	Mitsubishi	1
Arkansas	GMC	1
Arkansas	Chevrolet	1

Observations and Insights:

- Some states, like **Arizona** and **Arkansas**, have **multiple vehicle makers tied** with the same customer count.

- Chevrolet is the most preferred in **Alaska**, indicating its broad appeal even in less populous states.
- Brands like **Volkswagen, Suzuki, Mitsubishi, GMC** appear as top choices in some states, even with low counts.

Question 4: Find the overall average rating given by the customers. What is the average rating in each quarter?

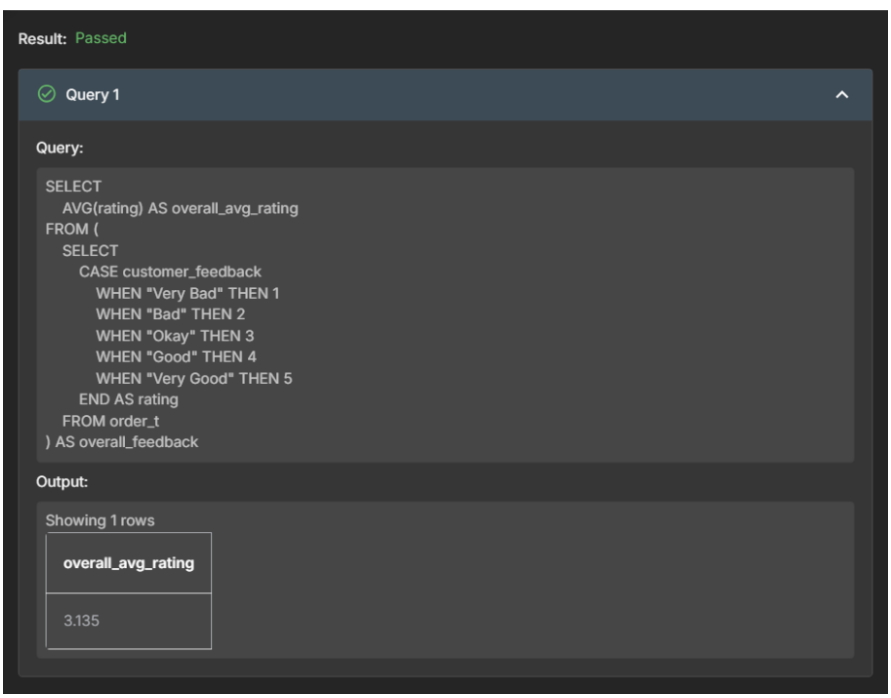
Consider the following mapping for ratings: "Very Bad": 1, "Bad": 2, "Okay": 3, "Good": 4, "Very Good": 5

4.1 Find the overall average rating given by the customers.

Solution Query:

```
SELECT AVG(rating) AS overall_avg_rating
FROM (
    SELECT
        CASE customer_feedback
            WHEN "Very Bad" THEN 1
            WHEN "Bad" THEN 2
            WHEN "Okay" THEN 3
            WHEN "Good" THEN 4
            WHEN "Very Good" THEN 5
        END AS rating
    FROM order_t
) AS overall_feedback;
```

Output:



Observations and Insights:

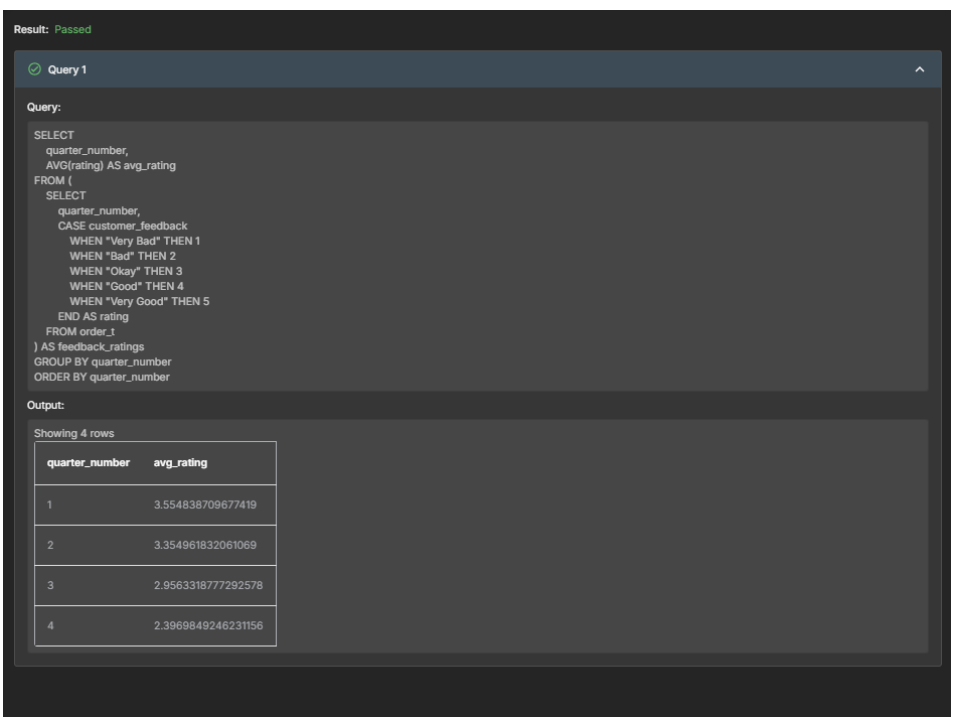
- The **average rating** is **3.135** means that most customers rated their experience as "Okay" or slightly better.

4.2 What is the average rating in each quarter?

Solution Query:

```
SELECT quarter_number, AVG(rating) AS avg_rating
FROM (
    SELECT
        quarter_number,
        CASE customer_feedback
            WHEN 'Very Bad' THEN 1
            WHEN 'Bad' THEN 2
            WHEN 'Okay' THEN 3
            WHEN 'Good' THEN 4
            WHEN 'Very Good' THEN 5
        END AS rating
    FROM order_t
) AS feedback_ratings
GROUP BY quarter_number
ORDER BY quarter_number;
```

Output:



quarter_number	avg_rating
1	3.554838709677419
2	3.354961832061069
3	2.9563318777292578
4	2.3969849246231156

Observations and Insights:

- **Customer ratings are going down each quarter** – starting from **3.55 in Q1** to **2.40 in Q4**.
- This means customer satisfaction is getting worse over time.
- The business should investigate what has changed and take steps to improve customer experience.

Question 5: Find the percentage distribution of feedback from the customers. Are customers getting more dissatisfied over time?

Solution Query:

```
SELECT quarter_number,
  ROUND(SUM(CASE customer_feedback WHEN 'Very Bad' THEN 1 ELSE 0 END) * 100.0 /
COUNT(*), 2) AS very_bad_percent,
  ROUND(SUM(CASE customer_feedback WHEN 'Bad' THEN 1 ELSE 0 END) * 100.0 /
COUNT(*), 2) AS bad_percent,
  ROUND(SUM(CASE customer_feedback WHEN 'Okay' THEN 1 ELSE 0 END) * 100.0 /
COUNT(*), 2) AS okay_percent,
  ROUND(SUM(CASE customer_feedback WHEN 'Good' THEN 1 ELSE 0 END) * 100.0 /
COUNT(*), 2) AS good_percent,
  ROUND(SUM(CASE customer_feedback WHEN 'Very Good' THEN 1 ELSE 0 END) * 100.0 /
COUNT(*), 2) AS very_good_percent

FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number;
```

Output:

Result: Passed

Query 1

Query:

```
SELECT
  quarter_number,
  ROUND(SUM(CASE customer_feedback WHEN 'Very Bad' THEN 1 ELSE 0 END) * 100.0 / COUNT(*), 2) AS very_bad_percent,
  ROUND(SUM(CASE customer_feedback WHEN 'Bad' THEN 1 ELSE 0 END) * 100.0 / COUNT(*), 2) AS bad_percent,
  ROUND(SUM(CASE customer_feedback WHEN 'Okay' THEN 1 ELSE 0 END) * 100.0 / COUNT(*), 2) AS okay_percent,
  ROUND(SUM(CASE customer_feedback WHEN 'Good' THEN 1 ELSE 0 END) * 100.0 / COUNT(*), 2) AS good_percent,
  ROUND(SUM(CASE customer_feedback WHEN 'Very Good' THEN 1 ELSE 0 END) * 100.0 / COUNT(*), 2) AS very_good_percent
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number
```

Output:

Showing 4 rows

quarter_number	very_bad_percent	bad_percent	okay_percent	good_percent	very_good_percent
1	10.97	11.29	19.03	28.71	30
2	14.89	14.12	20.23	22.14	28.63
3	17.9	22.71	21.83	20.96	16.59
4	30.65	29.15	20.1	10.05	10.05

\Observations and Insights:

- "Very Bad" ratings increased from **11% in Q1 to 31% in Q4**.
- "Very Good" ratings fell from **30% in Q1 to just 10% in Q4**.
- By **Q4, nearly 60%** of customers gave either "Very Bad" or "Bad" feedback, showing a clear decline in customer satisfaction.

Question 6: What is the trend of the number of orders by quarter?

Solution Query:

```
SELECT quarter_number, COUNT(DISTINCT order_id) order_count
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number;
```

Output:

Result: Passed

Query 1

Query:

```
SELECT quarter_number, COUNT(DISTINCT order_id) order_count
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number
```

Output:

Showing 4 rows

quarter_number	order_count
1	310
2	262
3	229
4	199

Observations and Insights:

- **Order numbers drop each quarter**, starting from **310 in Q1** to just **199 in Q4**.
- This shows a **steady decline in customer purchases** throughout the year.

Question 7: Calculate the net revenue generated by the company. What is the quarter-over-quarter % change in net revenue?

Solution Query:

```
SELECT quarter_number, SUM(quantity * (vehicle_price - discount)) AS net_revenue,
ROUND(
   100.0*(SUM(quantity * (vehicle_price - discount)) -
    LAG(SUM(quantity * (vehicle_price - discount))) OVER (ORDER BY
quarter_number)) /
    LAG(SUM(quantity * (vehicle_price - discount))) OVER (ORDER BY
quarter_number), 2) AS quarter_o_quarter_change_perc
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number;
```

Output:

Result: Passed

Query 1

Query:

```
SELECT
  quarter_number,
  SUM(quantity * (vehicle_price - discount)) AS net_revenue,
  ROUND(
   100.0*(SUM(quantity * (vehicle_price - discount)) -
    LAG(SUM(quantity * (vehicle_price - discount))) OVER (ORDER BY quarter_number)) /
    LAG(SUM(quantity * (vehicle_price - discount))) OVER (ORDER BY quarter_number),
  2) AS quarter_o_quarter_change_perc
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number
```

Output:

Showing 4 rows

quarter_number	net_revenue	quarter_o_quarter_cha...
1	39637378.160000026	
2	32913497.43999998	-16.96
3	29435188.489999995	-10.57
4	23495814.140000004	-20.18

Observations and Insights:

- The **net revenue** is decreasing every quarter, starting from around **39.6 million in Q1** to **23.5 million in Q4**.

- The biggest decline happened **in Q4**, with a **20.18% drop**, which is a major warning sign for the business.
- This steady decline in revenue suggests underlying problems, such as customer dissatisfaction, reduced orders, or poor product service.

Question 8: What is the trend of net revenue and orders by quarters?

Solution Query:

```
SELECT quarter_number,
    SUM(quantity * (vehicle_price - discount)) AS net_revenue,
    COUNT(order_id) AS total_orders
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number;
```

Output:

Result: Passed

Query 1

Query:

```
SELECT
    quarter_number,
    SUM(quantity * (vehicle_price - discount)) AS net_revenue,
    COUNT(order_id) AS total_orders
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number
```

Output:

Showing 4 rows

quarter_number	net_revenue	total_orders
1	39637378.160000026	310
2	32913497.43999998	262
3	29435188.489999995	229
4	23495814.140000004	199

Observations and Insights:

- Both **revenue** and **orders** are decreasing every quarter, which shows a consistent downward trend.

- From Q1 to Q4, orders dropped by **36%**, and revenue dropped by over **40%**, which is a significant decline.
- The decline in orders is directly impacting on the drop in revenue, suggesting lower customer engagement or satisfaction.

Question 9: What is the average discount offered for different types of credit cards?

Solution Query:

```
SELECT credit_card_type, ROUND(AVG(discount), 2) AS Average_Discount
FROM customer_t
JOIN order_t USING (customer_id)
GROUP BY credit_card_type
ORDER BY Average_Discount DESC;
```

Output:

Result: Passed

Query 1

Query:

```
SELECT credit_card_type, ROUND(AVG(discount), 2) AS Average_Discount
FROM customer_t
JOIN order_t USING (customer_id)
GROUP BY credit_card_type
ORDER BY Average_Discount DESC
```

Output:

Showing first 10 rows out of 16 rows

credit_card_type	Average_Discount
laser	0.64
mastercard	0.63
visa-electron	0.62
maestro	0.62
instapayment	0.62

Observations and Insights:

- Laser card holders receive the highest average discount at **0.64**, followed closely by Mastercard users at **0.63**.

- Several card types like Visa Electron, Maestro, and Instapaymet all received an average discount of **0.62,** showing that there was little difference.
- The discounts offered by various credit card kinds are fairly comparable, indicating that the discount policy is generally consistent and not card specific.

Question 10: What is the average time taken to ship the placed orders for each quarter?

Solution Query:

```
SELECT quarter_number,
    ROUND(AVG(julianday(ship_date) - julianday(order_date)), 2) AS
avg_shipping_days
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number;
```

Output:

Result: Passed

Query 1

Query:

```
SELECT quarter_number,
    ROUND(AVG(julianday(ship_date) - julianday(order_date)), 2) AS avg_shipping_days
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number
```

Output:

Showing 4 rows

quarter_number	avg_shipping_days
1	57.17
2	71.11
3	117.76
4	174.1

Observations and Insights:

- The average shipping time is **increasing every quarter**, which means orders are taking longer to deliver overtime.

- This growing delay could lead to customer dissatisfaction, possibly explaining the drop in ratings and orders seen in other data.

Business Metrics Overview

Total Revenue	Total Orders	Total Customers	Average Rating
125481878.23	1000	994	3.14
Last Quarter Revenue	**Last quarter Orders**	**Average Days to Ship**	**% Good Feedback**
23495814.14	199	97.96	44.10

Business Recommendations

- Consider using predictive tools to allocate resources more efficiently based on demand trends.
- Prioritize common complaints and implement a resolution plan.
- Offer discounts, free shipping, or loyalty rewards for repeat customers to rebuild trust.
- Implement dashboards to track these metrics in real time.
- Analyze order-to-ship time per region or product type to identify limitations.